

ELE2038 Control Coursework

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1 Introduction

second-order terms gives

2 System Linearisation

$$m_{\text{tot}}\ddot{x}_A + b_x\dot{x}_A + m_2L\ddot{\theta} = \tilde{F}, \quad (5)$$

2.1 Equilibrium Point

and similarly, substituting into Equation (??) yields

To design a PID controller, the nonlinear system dynamics of Equations (??) and (??) must first be linearised about an appropriate equilibrium point. An equilibrium is a state in which the system remains at rest indefinitely, i.e., all velocities and accelerations are zero. Setting $\dot{x}_A = \ddot{x}_A = \dot{\theta} = \ddot{\theta} = 0$ and substituting into Equation (??) gives $F = 0$. Substituting into Equation (??) yields

$$m_2L\ddot{x}_A + m_2L^2\ddot{\theta} + b_\theta\dot{\theta} + m_2gL\tilde{\theta} = 0, \quad (6)$$

where $m_{\text{tot}} = m_1 + m_2$.

$$m_2gL \sin \theta = 0, \quad (1)$$

which implies $\theta^* = 0$. The equilibrium point is therefore

$$x_A^* = x_A^{\text{ref}}, \quad \theta^* = 0, \quad \dot{x}_A^* = 0, \quad \dot{\theta}^* = 0, \quad F^* = 0, \quad (2)$$

which corresponds physically to the cable hanging vertically at rest.

3 Transfer Function Derivation

3.1 Laplace Transform of Linearised Equations

Taking the Laplace transform of Equations (5) and (6), assuming zero initial conditions, gives

$$m_{\text{tot}}s^2\tilde{X}_A + b_xs\tilde{X}_A + m_2Ls^2\tilde{\Theta} = \tilde{F}, \quad (7)$$

$$m_2Ls^2\tilde{X}_A + m_2L^2s^2\tilde{\Theta} + b_\theta s\tilde{\Theta} + m_2gL\tilde{\Theta} = 0. \quad (8)$$

2.2 Small Perturbation Analysis

We introduce small perturbations about the equilibrium,

$$x_A = x_A^{\text{ref}} + \tilde{x}_A, \quad \theta = \tilde{\theta}, \quad F = \tilde{F}, \quad (3)$$

where tildes denote small deviations. For small $\tilde{\theta}$, Taylor's theorem gives the first-order approximations

$$\sin \tilde{\theta} \approx \tilde{\theta}, \quad \cos \tilde{\theta} \approx 1, \quad \dot{\theta}^2 \approx 0. \quad (4)$$

3.2 Solving for $\tilde{\Theta}$

Rearranging Equation (8) to express $\tilde{X}_A$ in terms of $\tilde{\Theta}$,

$$\tilde{X}_A = \frac{-(m_2L^2s^2 + b_\theta s + m_2gL)}{m_2Ls^2} \tilde{\Theta}. \quad (9)$$

Substituting Equation (9) into Equation (7) and multiplying through by m_2Ls^2 gives

2.3 Linearised Equations of Motion

Substituting the perturbations and small-angle approximations into Equation (??) and neglecting all

$$\begin{aligned} & -m_{\text{tot}}s^2(m_2L^2s^2 + b_\theta s + m_2gL)\tilde{\Theta} \\ & -b_xs(m_2L^2s^2 + b_\theta s + m_2gL)\tilde{\Theta} + m_2^2L^2s^4\tilde{\Theta} = m_2Ls^2\tilde{F}. \end{aligned} \quad (10)$$

Collecting terms by powers of s and noting that $m_2^2 L^2 - m_{\text{tot}} m_2 L^2 = -m_1 m_2 L^2$, the left-hand side becomes

$$\tilde{\Theta}[-m_1 m_2 L^2 s^4 - (m_{\text{tot}} b_\theta + b_x m_2 L^2) s^3 - (m_{\text{tot}} m_2 g L + b_x b_\theta) s^2 - b_x m_2 g L s] = m_2 L s^2 \tilde{F}. \quad (11)$$

Factoring $-s$ from the left-hand side and solving for $\tilde{\Theta}$,

$$\tilde{\Theta}(s) = \frac{-m_2 L s}{D(s)} \tilde{F}(s), \quad (12)$$

where the denominator polynomial is defined as

$$D(s) = a_3 s^3 + a_2 s^2 + a_1 s + a_0, \quad (13)$$

with coefficients $a_3 = m_1 m_2 L^2$, $a_2 = b_x m_2 L^2 + b_\theta m_{\text{tot}}$, $a_1 = m_2 g L m_{\text{tot}} + b_x b_\theta$, and $a_0 = m_2 b_x g L$.

3.3 Solving for $\tilde{X}_A$

Substituting Equation (12) into Equation (9) gives

$$\tilde{X}_A(s) = \frac{m_2 L^2 s^2 + b_\theta s + m_2 g L}{s D(s)} \tilde{F}(s). \quad (14)$$

3.4 Hook Position Transfer Function

From the crane geometry, the hook position satisfies $x_B = x_A + L \sin \theta$. Linearising using $\sin \theta \approx \theta$ gives $\tilde{x}_B = \tilde{x}_A + L \tilde{\theta}$, or in the Laplace domain,

$$\tilde{X}_B(s) = \tilde{X}_A(s) + L \tilde{\Theta}(s). \quad (15)$$

Substituting Equations (12) and (14),

$$\begin{aligned} \tilde{X}_B(s) &= \frac{m_2 L^2 s^2 + b_\theta s + m_2 g L}{s D(s)} \tilde{F} + L \cdot \frac{-m_2 L s}{D(s)} \tilde{F} \\ &= \frac{(m_2 L^2 s^2 + b_\theta s + m_2 g L) - m_2 L^2 s^2}{s D(s)} \tilde{F}, \end{aligned} \quad (16)$$

which simplifies to

$$\frac{\tilde{X}_B(s)}{\tilde{F}(s)} = \frac{b_\theta s + m_2 g L}{s(a_3 s^3 + a_2 s^2 + a_1 s + a_0)}, \quad (17)$$

confirming the expected form of Equation (3) in the problem statement, with the coefficients as defined above.

4	Transfer Function
5	Block Diagram
6	PID Controller Design
7	Simulation Results
8	Robustness Analysis